

WEEK 2 HOMEWORK

Question 4.1

Describe a situation or problem from your job, everyday life, current events, etc., for which a clustering model would be appropriate. List some (up to 5) predictors that you might use.

Question 4.2

The *iris* data set `iris.txt` contains 150 data points, each with four predictor variables and one categorical response. The predictors are the width and length of the sepal and petal of flowers and the response is the type of flower. The data is available from the R library `datasets` and can be accessed with `iris` once the library is loaded. It is also available at the UCI Machine Learning Repository (<https://archive.ics.uci.edu/ml/datasets/Iris>). *The response values are only given to see how well a specific method performed and should not be used to build the model.*

Use the R function `kmeans` to cluster the points as well as possible. Report the best combination of predictors, your suggested value of k , and how well your best clustering predicts flower type.

Question 5.1

Using crime data from the file `uscrime.txt` (<http://www.statsci.org/data/general/uscrime.txt>, description at <http://www.statsci.org/data/general/uscrime.html>), test to see whether there are any outliers in the last column (number of crimes per 100,000 people). Use the `grubbs.test` function in the `outliers` package in R.

Question 6.1

Describe a situation or problem from your job, everyday life, current events, etc., for which a Change Detection model would be appropriate. Applying the CUSUM technique, how would you choose the critical value and the threshold?

Question 6.2

1. Using July through October daily-high-temperature data for Atlanta for 1996 through 2015, use a CUSUM approach to identify when unofficial summer ends (i.e., when the weather starts cooling off) each year. You can get the data that you need from the file `temps.txt` or online, for example at <http://www.iweather.net/atlanta-weather-records> or <https://www.wunderground.com/history/airport/KFTY/2015/7/1/CustomHistory.html>. You can use R if you'd like, but it's straightforward enough that an Excel spreadsheet can easily do the job too.
2. Use a CUSUM approach to make a judgment of whether Atlanta's summer climate has gotten warmer in that time (and if so, when).
3. Get your favorite AI to repeat part 2; compare its answer with yours, and judge where each one is better and worse, and (if at all) where each one is wrong. Include the prompt/chat logs in your homework submission, and specify which AI you used.

- a. **For this part of the question only**, the course AI Use Policy is overridden; for this part, I want you to see what solution the AI gives. For suggestions on how to effectively use AI as a productivity tool for this sort of thing, please see the AI Use Policy (which includes a tutorial on how to use AI) in the Course Information module on Canvas (for Georgia Tech students) or edX (for edX students).